

IIN 3502 Distributed Network

Lab 03

Index No. – ICT/23/810

Name – A.K. Nipuna Darshana

Questions & Answers

01. If Switch2 receives traffic tagged for VLAN 10 passing from Switch3 to Switch0, why will Switch2 drop that traffic if VLAN 10 is not manually created on Switch2?

- Because Switch 2 is in transparent mode, Transparent switch only do VTP advertisements forward and not collect for a VLAN database, then if not configured correctly switch 2 By manually, Switch 2 dropped traffic of vlan 10

02. What triggers the VTP Configuration Revision Number to increment on the VTP Server, and what happens when a client switch receives an advertisement with a higher revision number?

- Whenever a change (such as adding, deleting, or renaming a VLAN) is made to the VLAN database on a VTP server, the revision number increases by one. If a client receives an advertisement with a higher revision number than its own, it discards its current VLAN database and overwrites it with the new database received from the server.

03. What is the specific purpose of the encapsulation dot1Q <vlan-id> command configured on the sub interfaces of Router0?

- That command tells the router, "This is the VLAN ID associated with data arriving via this sub-interface." This enables the router to separate tagged traffic coming through a single physical interface and route it to the appropriate VLAN. (This is the basis of the "Router-on-a-Stick" configuration).

04. If PC0 and PC1 on Switch1 cannot ping their default gateway on Router0, list three potential layer 2 configuration errors you should check on Switch1 or Switch0.

- Trunk Port Error: The link between Switch0 and Switch1 has not been configured as a trunk (or only one side has been configured as a trunk).

- VLAN Allowed Error: VLAN 10 has not been permitted on the trunk port (the “switchport trunk allowed vlan 10” command was not entered).

- Access Port Error: The port connected to PC0 has been mistakenly assigned to a different VLAN, or the port has not been set to Access Mode.

- Native VLAN Mismatch: The Native VLANs on the two ends of the trunk do not match.